

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy as sp

%matplotlib inline
```

```
amazon_dataset = pd.read_csv("/content/amazon.csv")
```

```
print(amazon_dataset.head())
```

```
3      ₹699      53%    4.2    94,363
4      ₹399      61%    4.2    16,905
```

```
about_product \
0 High Compatibility : Compatible With iPhone 12...
1 Compatible with all Type C enabled devices, be...
2 【 Fast Charger& Data Sync】 -With built-in safet...
3 The boAt Deuce USB 300 2 in 1 cable is compati...
4 [CHARGE & SYNC FUNCTION]- This cable comes wit...
```

```
user_id \
0 AG3D604STAQKAY2UVGEUV46KN35Q,AHMY5CWJMMK5BJRBB...
1 AECPFYFQVRUWC3KGNLJIOREFP5LQ,AGYYVPDD7YG7FYNBX...
2 AGU3BBQ2V2DDAMOAKGFAWDDQ6QHA,AESFLDV2PT363T2AQ...
3 AEWAZDZZJLQUYVVOVGBEUKSLXHQ5A,AG5HTSFRRE6NL3M5S...
4 AE3Q6KSUK5P75D5HFYHCRAOLODSA,AFUGIFH5ZAFXRDSZH...
```

```
user_name \
0 Manav,Adarsh gupta,Sundeep,S.Sayeed Ahmed,jasp...
1 ArdKn,Nirbhay kumar,Sagar Viswanathan,Asp,Plac...
2 Kunal,Himanshu,viswanath,sai niharka,saqib mal...
3 Omkar dhale,JD,HEMALATHA,Ajwadh a.,amar singh ...
4 rahuls6099,Swasat Borah,Ajay Wadke,Pranali,RVK...
```

```
review_id \
0 R3HXWT0LRP0NMF,R2AJM3LFTLZHFO,R6AQJGUP6P86,R1K...
1 RGIQEG07R9HS2,R1SMWZQ86XIN8U,R2J3Y1WL29GWDE,RY...
2 R3J3EQQ9TZI5ZJ,R3E7WBGK7ID0KV,RWU79XKQ6I1QF,R2...
3 R3EEUZKKK9J36I,R3HJVYCLYOY554,REDECAZ7AMPQC,R1...
4 R1BP4L2HH9TFUP,R16PVJEXKV6QZS,R2UPDB81N66T4P,R...
```

```
review_title \
0 Satisfied,Charging is really fast,Value for mo...
1 A Good Braided Cable for Your Type C Device,Go...
2 Good speed for earlier versions,Good Product,W...
3 Good product,Good one,Nice,Really nice product...
4 As good as original,Decent,Good one for second...
```

```
review_content \
0 Looks durable Charging is fine tooNo complains...
1 I ordered this cable to connect my phone to An...
2 Not quite durable and sturdy,https://m.media-a......
3 Good product,long wire,Charges good,Nice,I bou...
4 Bought this instead of original apple, does th...
```

```
img_link \
0 https://m.media-amazon.com/images/W/WEBP\_40237......
1 https://m.media-amazon.com/images/W/WEBP\_40237......
2 https://m.media-amazon.com/images/W/WEBP\_40237......
3 https://m.media-amazon.com/images/I/41V5FtEWPk......
4 https://m.media-amazon.com/images/W/WEBP\_40237......
```

```
product_link
0 https://www.amazon.in/Wayona-Braided-WN3LG1-Sy......
1 https://www.amazon.in/Ambrane-Unbreakable-Char......
2 https://www.amazon.in/Sounce-iPhone-Charging-C......
3 https://www.amazon.in/Deuce-300-Resistant-Tang......
4 https://www.amazon.in/Portronics-Konnnect-POR-1......
```

```
display(amazon_dataset)
```

	product_id	product_name	category	discounted_price	actual_price	discount_perce
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories Accessories&Peripherals ...	₹399	₹1,099	
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories Accessories&Peripherals ...	₹199	₹349	
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories Accessories&Peripherals ...	₹199	₹1,899	
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories Accessories&Peripherals ...	₹329	₹699	
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories Accessories&Peripherals ...	₹154	₹399	
...	...	...	...	...	...	...
1460	B08L7J3T31	Noir Aqua - 5pcs PP Spun Filter + 1 Spanner ...	Home&Kitchen Kitchen&HomeAppliances WaterPurif...	₹379	₹919	
1461	B01M6453MB	Prestige Delight PRWO Electric Rice Cooker (1 ...	Home&Kitchen Kitchen&HomeAppliances SmallKitch...	₹2,280	₹3,045	
1462	B009P2LIL4	Bajaj Majesty RX10 2000 Watts Heat Convector R...	Home&Kitchen Heating,Cooling&AirQuality RoomHe...	₹2,219	₹3,080	
1463	B00J5DYCCA	Havells Ventil Air DSP 230mm Exhaust Fan (Pist...	Home&Kitchen Heating,Cooling&AirQuality Fans E...	₹1,399	₹1,890	
1464	B01486F4G6	Borosil Jumbo 1000-Watt Grill Sandwich Maker (...	Home&Kitchen Kitchen&HomeAppliances SmallKitch...	₹2,863	₹3,690	

1465 rows × 16 columns

Next steps: [Generate code with amazon_dataset](#) [New interactive sheet](#)

amazon_dataset.shape

(1465, 16)

amazon_dataset.columns

```
Index(['product_id', 'product_name', 'category', 'discounted_price',
      'actual_price', 'discount_percentage', 'rating', 'rating_count',
      'about_product', 'user_id', 'user_name', 'review_id', 'review_title',
      'review_content', 'img_link', 'product_link'],
      dtype='object')
```

amazon_dataset.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1465 entries, 0 to 1464
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---
```

```
--- -----  
0 product_id      1465 non-null object  
1 product_name    1465 non-null object  
2 category        1465 non-null object  
3 discounted_price 1465 non-null object  
4 actual_price    1465 non-null object  
5 discount_percentage 1465 non-null object  
6 rating          1465 non-null object  
7 rating_count    1463 non-null object  
8 about_product   1465 non-null object  
9 user_id         1465 non-null object  
10 user_name      1465 non-null object  
11 review_id      1465 non-null object  
12 review_title   1465 non-null object  
13 review_content 1465 non-null object  
14 img_link       1465 non-null object  
15 product_link   1465 non-null object  
dtypes: object(16)  
memory usage: 183.3+ KB
```

```
amazon_dataset.isnull().sum()
```

	0
product_id	0
product_name	0
category	0
discounted_price	0
actual_price	0
discount_percentage	0
rating	0
rating_count	2
about_product	0
user_id	0
user_name	0
review_id	0
review_title	0
review_content	0
img_link	0
product_link	0

dtype: int64

```
pd.set_option('display.max_columns', None)
```

```
amazon_dataset.head(5)
```

	product_id	product_name	category	discounted_price	actual_price	discount_percentage
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories Accessories&Peripherals ...	₹399	₹1,099	64%
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories Accessories&Peripherals ...	₹199	₹349	43%
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories Accessories&Peripherals ...	₹199	₹1,899	90%
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories Accessories&Peripherals ...	₹329	₹699	53%
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories Accessories&Peripherals ...	₹154	₹399	61%

Next steps:

[Generate code with amazon_dataset](#)

[New interactive sheet](#)

```
amazon_dataset['discounted_price'] = amazon_dataset['discounted_price'].str.replace('₹', '')
amazon_dataset['discounted_price'] = amazon_dataset['discounted_price'].str.replace(',', '')
amazon_dataset['discounted_price'] = amazon_dataset['discounted_price'].astype(float)

amazon_dataset['actual_price'] = amazon_dataset['actual_price'].str.replace('₹', '')
amazon_dataset['actual_price'] = amazon_dataset['actual_price'].str.replace(',', '')
amazon_dataset['actual_price'] = amazon_dataset['actual_price'].astype(float)
```

```
amazon_dataset['discount_percentage'] = amazon_dataset['discount_percentage'].str.replace('%', '').astype(float)
amazon_dataset['discount_percentage'] = amazon_dataset['discount_percentage'] / 100
```

```
amazon_dataset['rating'].value_counts()
```

count	
rating	
4.1	244
4.3	230
4.2	228
4.0	129
3.9	123
4.4	123
3.8	86
4.5	75
4	52
3.7	42
3.6	35
3.5	26
4.6	17
3.3	16
3.4	10
4.7	6
3.1	4
3.0	3
4.8	3
5.0	3
2.8	2
3.2	2
2.3	1
	1
2	1
3	1
2.6	1
2.9	1

dtype: int64

```
amazon_dataset.query('rating == "|")
```

product_id		product_name	category	discounted_price	actual_price	discount_percent
1279	B08L12N5H1	Eureka Forbes	Home&Kitchen Kitchen&HomeAppliances Vacuum,Cle...	2099.0	2499.0	
		car Vac 100				
		Watts				
		Powerful Sucti...				

```
amazon_dataset['rating'] = amazon_dataset['rating'].str.replace('|', '3.9').astype(float)
```

```
amazon_dataset['rating_count'] =amazon_dataset['rating_count'].str.replace(',','').astype(float)
```

```
amazon_dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1465 entries, 0 to 1464
Data columns (total 16 columns):
#   Column              Non-Null Count  Dtype
---  -
#   Column              Non-Null Count  Dtype
```

```
0 product_id      1465 non-null object
1 product_name    1465 non-null object
2 category        1465 non-null object
3 discounted_price 1465 non-null float64
4 actual_price    1465 non-null float64
5 discount_percentage 1465 non-null float64
6 rating          1465 non-null float64
7 rating_count    1463 non-null float64
8 about_product   1465 non-null object
9 user_id         1465 non-null object
10 user_name      1465 non-null object
11 review_id      1465 non-null object
12 review_title   1465 non-null object
13 review_content 1465 non-null object
14 img_link       1465 non-null object
15 product_link   1465 non-null object
dtypes: float64(5), object(11)
memory usage: 183.3+ KB
```

```
amazon_dataset.describe()
```

	discounted_price	actual_price	discount_percentage	rating	rating_count	
count	1465.000000	1465.000000	1465.000000	1465.000000	1463.000000	
mean	3125.310874	5444.990635	0.476915	4.096451	18295.541353	
std	6944.304394	10874.826864	0.216359	0.291620	42753.864952	
min	39.000000	39.000000	0.000000	2.000000	2.000000	
25%	325.000000	800.000000	0.320000	4.000000	1186.000000	
50%	799.000000	1650.000000	0.500000	4.100000	5179.000000	
75%	1999.000000	4295.000000	0.630000	4.300000	17336.500000	
max	77990.000000	139900.000000	0.940000	5.000000	426973.000000	

```
amazon_dataset.isnull().sum()
```

	0
product_id	0
product_name	0
category	0
discounted_price	0
actual_price	0
discount_percentage	0
rating	0
rating_count	2
about_product	0
user_id	0
user_name	0
review_id	0
review_title	0
review_content	0
img_link	0
product_link	0

```
dtype: int64
```

```
amazon_dataset['rating_count'] = amazon_dataset['rating_count'].fillna(value=amazon_dataset['rating_count'].median())
```

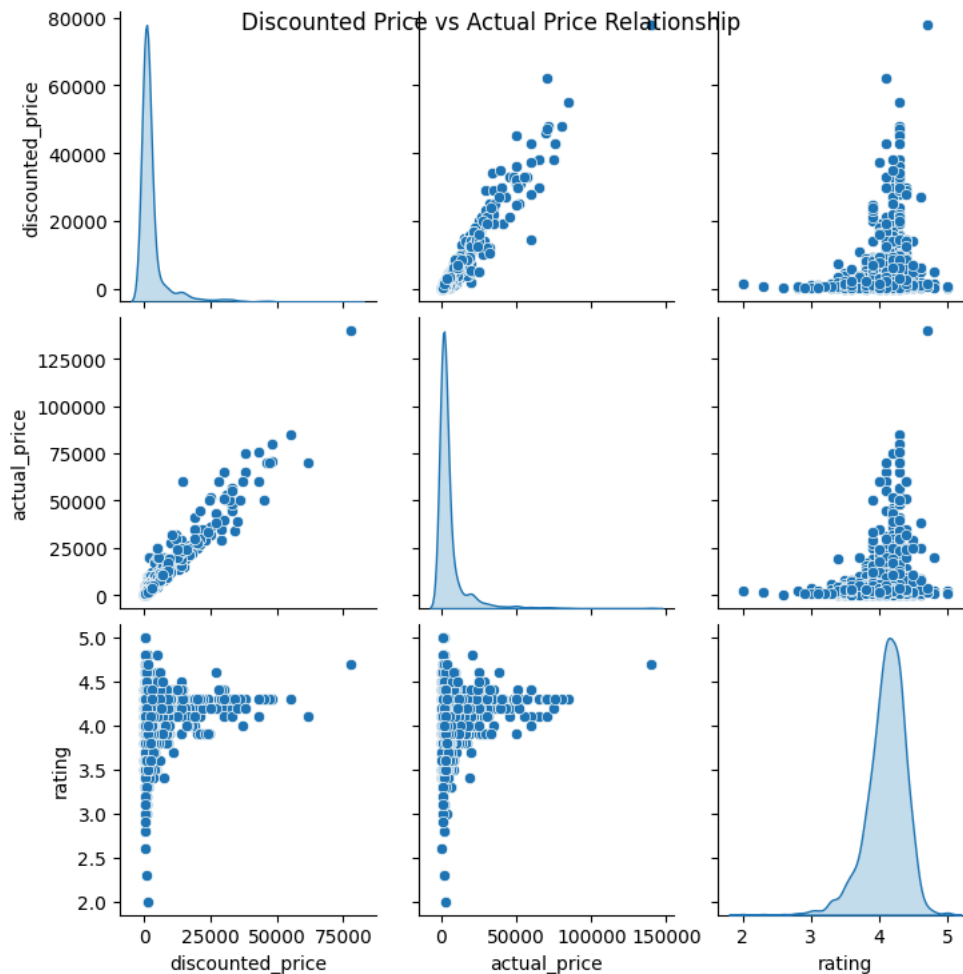
```
any_dubliactes = amazon_dataset.duplicated(subset=['product_id', 'product_name', 'category', 'discounted_price',
'actual_price', 'discount_percentage', 'rating', 'rating_count',
'about_product', 'user_id', 'user_name', 'review_id', 'review_title',
'review_content', 'img_link', 'product_link']).any()
```

```
any_dubliactes
```

```
np.False_
```

```
sns.set_style('whitegrid')
plt.figure(figsize=(15, 10))
```

```
sns.pairplot(amazon_dataset[['discounted_price', 'actual_price', 'rating']], diag_kind="kde")
plt.suptitle("Discounted Price vs Actual Price Relationship")
plt.show()
```



```
feature_cols = ['discounted_price', 'actual_price', 'discount_percentage', 'rating', 'rating_count']
corr = amazon_dataset[feature_cols].corr(numeric_only=True)
plt.figure(figsize=(10, 8))
sns.heatmap(corr, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Matrix')
plt.show()
```

